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5. Write a program for Implementation of K-Nearest Neighbours (K-NN) in Python.

1.

Study Hours	Attendance	Result
2	60	Fail
3	65	Fail
5	75	Pass
6	80	Pass
8	90	Pass

PROGRAM

```
import pandas as pd
from sklearn.neighbors import KNeighborsClassifier
df = pd.read_csv("student.csv")
X = df[["Study Hours", "Attendance"]]
y = df["Result"]
model = KNeighborsClassifier(n_neighbors=3)
model.fit(X, y)
= [[7, 85]]
print("Prediction:", model.predict(sample)[0])
```

OUTPUT

Prediction: Pass

2.

Weight	Sweetness	Fruit
120	70	Orange
125	72	Orange
145	90	Apple
150	92	Apple
160	88	Apple

PROGRAM

```
import pandas as pd
from sklearn.neighbors import KNeighborsClassifier
If = pd.read_csv("fruit.csv")X = df[["Weight", "Sweetness"]]
y = df["Fruit"]
model = KNeighborsClassifier(n_neighbors=3)
model.fit(X, y)
sample = [[148, 91]]
print("Prediction:", model.predict(sample)[0])
```

OUTPUT

Prediction: Apple

3.

Income	Credit Score	Loan
3	580	No
4	620	No
6	700	Yes
7	750	Yes

9	800	Yes
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PROGRAM

```
import pandas as pd
from sklearn.neighbors import KNeighborsClassifier
df = pd.read_csv("loan.csv")
X = df[["Income", "Credit Score"]]
y = df["Loan"]
model = KNeighborsClassifier(n_neighbors=3)
model.fit(X, y)
sample = [[8, 770]]
print("Prediction:", model.predict(sample)[0])
```

OUTPUT

Prediction: Yes

4.

Sepal Length	Petal Length	Flower
5.1	1.4	Setosa
4.9	1.4	Setosa
6.0	4.5	Versicolor
5.9	4.8	Versicolor
6.5	5.5	Virginica
6.7	5.6	Virginica

PROGRAM

```
import pandas as pd
from sklearn.neighbors import KNeighborsClassifier
df = pd.read_csv("iris.csv")
```

```
X = df[["Sepal Length", "Petal Length"]]
y = df["Flower"]
model = KNeighborsClassifier(n_neighbors=3)
model.fit(X, y)
sample = [[6.4, 5.4]]
print("Prediction:", model.predict(sample)[0])
```

OUTPUT

Prediction: Virginica

5.

Experience	Performance	Promotion
2	60	No
3	65	No
5	75	Yes
7	85	Yes
9	95	Yes

PROGRAM

```
import pandas as pd
from sklearn.neighbors import KNeighborsClassifier
df = pd.read_csv("employee.csv")
X = df[["Experience", "Performance"]]
y = df["Promotion"]
model = KNeighborsClassifier(n_neighbors=3)
model.fit(X, y)
sample = [[6, 80]]
print("Prediction:", model.predict(sample)[0])
```

OUTPUT

Prediction: Yes